

Deep Reinforcement Learning

Brief Introduction to Deep Learning

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Content

Content

Content:

- The three main learning paradigms
- Supervised learning
 - The learning diagram
 - Generalization
 - Bias & variance
 - Cross-validation
- Deep neural networks
 - Artificial neural networks
 - Multi-layer perceptron
 - Training
 - Regularization
 - Neural network architectures
- Linear regression as a special case

Useful references:

- Pattern recognition and machine learning ([Bishop 2006](#))
- Deep learning ([Bengio et al. 2017](#))
- The elements of statistical learning ([Hastie 2009](#))
- Learning from data: a short course ([Abu-Mostafa, Magdon-Ismail, and Lin 2012](#))

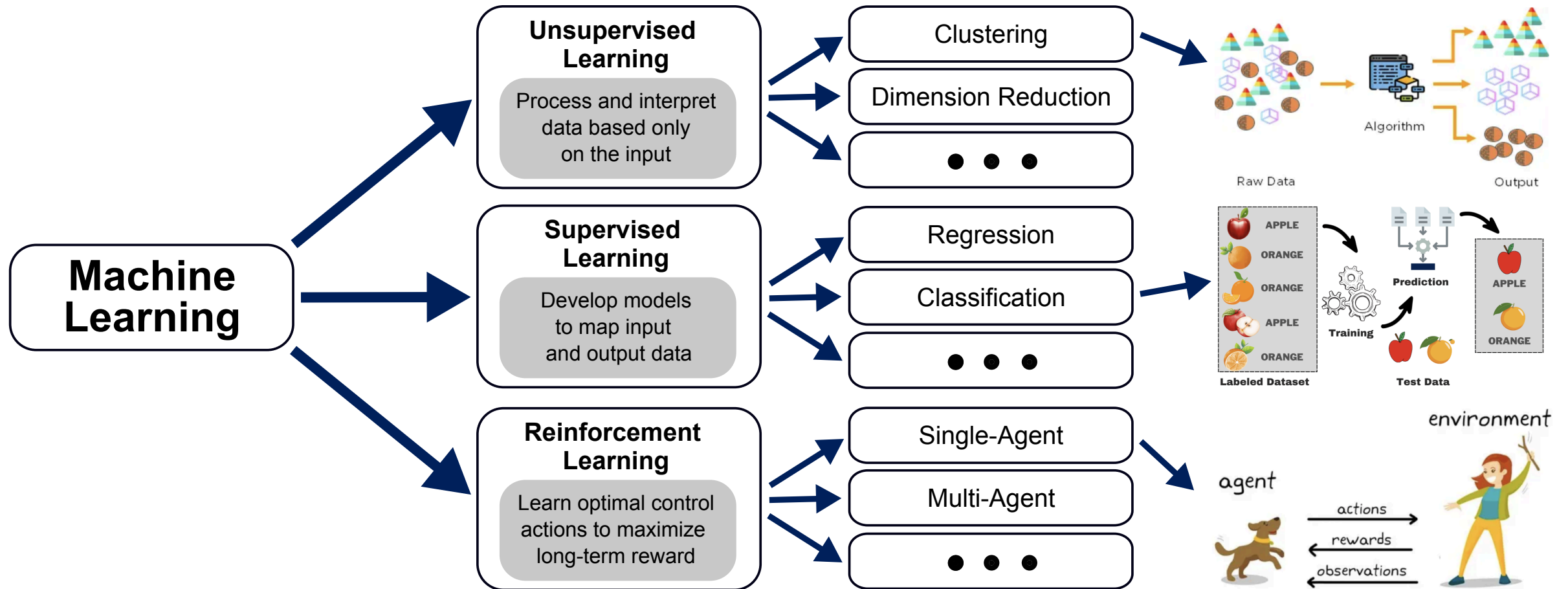
Where are we?

Chapter	Topic	Content
Basics & tabular methods		
1	Multi-armed bandits	Exploration-exploitation dilemma
2	Markov decision processes	Dynamics, rewards, policies
3	Dynamic programming	Optimal decision making with <i>full knowledge</i>
4	Monte Carlo methods	<i>Data-driven learning</i> from entire episodes
5	Temporal difference learning & Q-learning	<i>Data-driven learning</i> from individual transitions
Deep-learning-based methods		
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	
8	Deep Q-learning	
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10	Actor-critic algorithms	
11	Advanced algorithms	
Model-Based Control		

Chapter	Topic	Content
Advanced Topics		
<i>Lecture contents</i>		

The three main learning paradigms

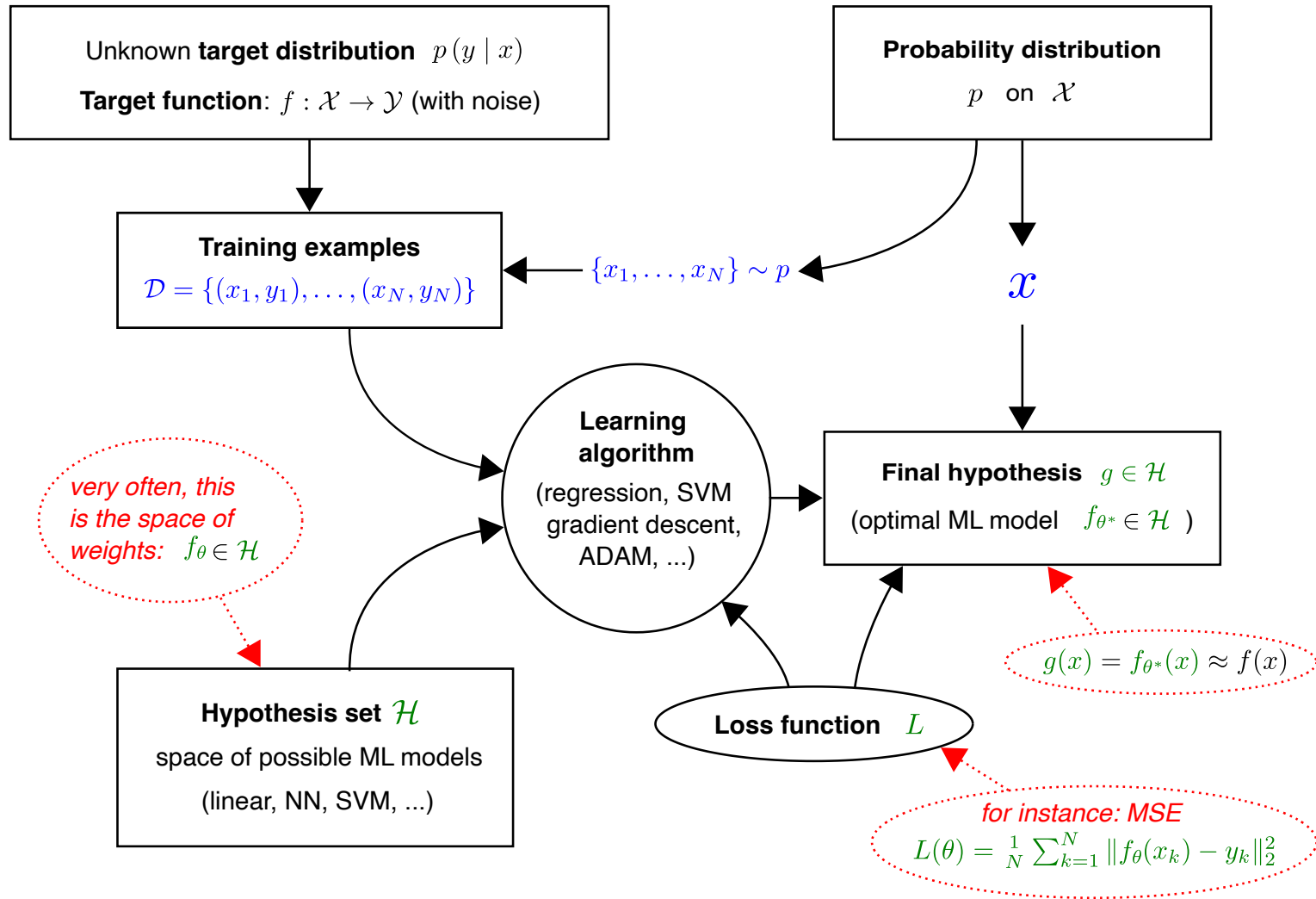
The three main learning paradigms



The three big learning paradigms (Abdelwanis et al. 2026) (Examples: *Unsupervised*, *Supervised*, *RL*)

Supervised learning

The learning diagram



The learning diagram (Abu-Mostafa, Magdon-Ismail, and Lin 2012)

Generalization

- What does it mean to have a perfect model on your training dataset $\mathcal{D}_{\text{train}}$, i.e., $L(\theta; \mathcal{D}_{\text{train}}) = 0$?
 $\Rightarrow$ we have simply memorized the data!
- Learning means that we get **good predictions on unseen data**:

$$\underbrace{L(\theta; \mathcal{D}_{\text{test}})}_{\text{in-sample error}} \approx \underbrace{L(\theta; \mathcal{D}_{\text{train}})}_{\text{out-of-sample error}} .$$

Supervised learning

Given a labeled (and probably noisy) dataset $\mathcal{D}_{\text{train}} = \{(x_1, y_1), \dots, (x_N, y_N)\}$, approximate the unknown mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ by a parametrizable ML model $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, such that

$$f_\theta(x_k) = \hat{y}_k \approx y_k \quad \forall (x_k, y_k) \in \mathcal{D}_{\text{test}}.$$

- **Goodness of fit** can be measured via many different metrics (e.g., mean squared error, classification accuracy, etc.).
- The dimension d of model parameters $\theta \in \mathbb{R}^d$ is adjustable in many model families, which trades off **bias** with **variance** (among other factors, leading to so-called under- and overfitting).

- On top of θ , an ML model might also have **hyperparameters** that can be optimized (e.g., number of layers in a neural network).

Bias-variance tradeoff (1)

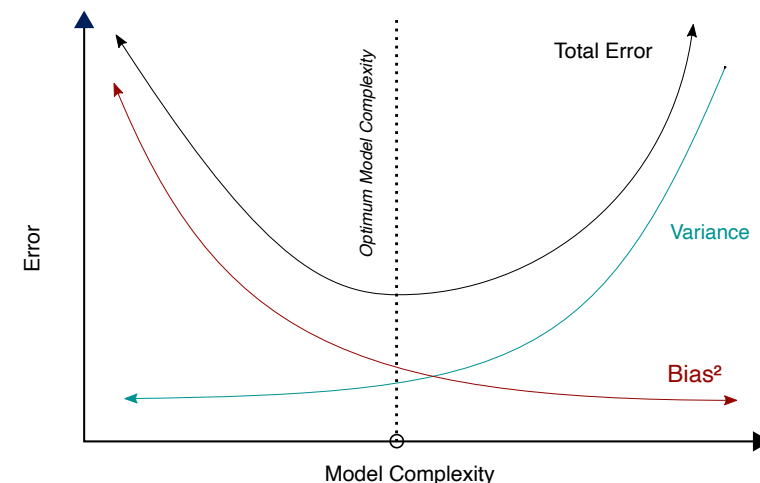
- In the ML context, **bias** denotes the error of the *average* model $\bar{f}_\theta$ when repeating the training with different datasets $\mathcal{D}_{\text{train},1}, \mathcal{D}_{\text{train},2}, \dots$:

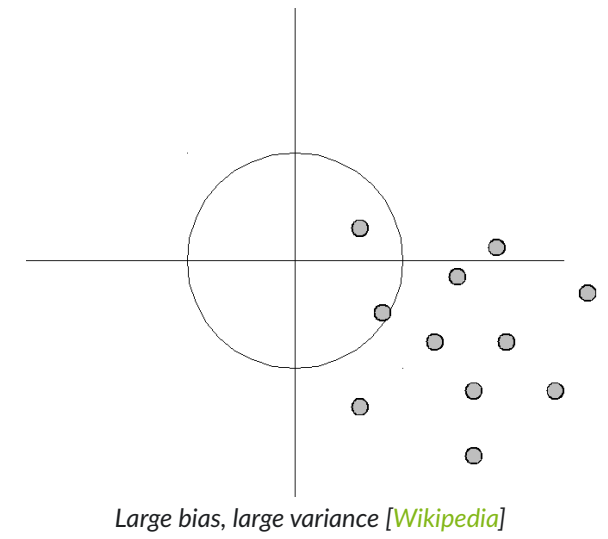
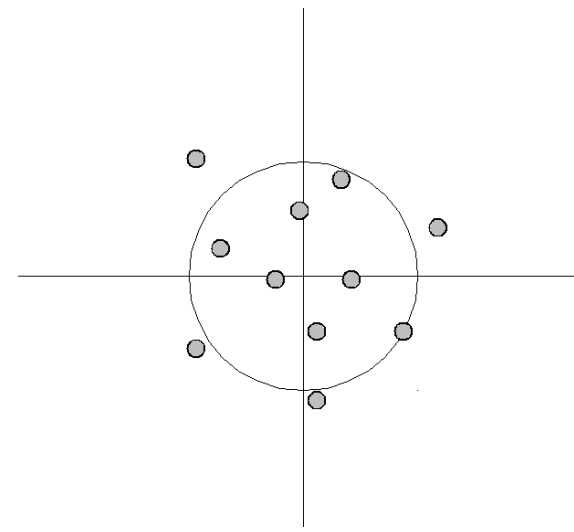
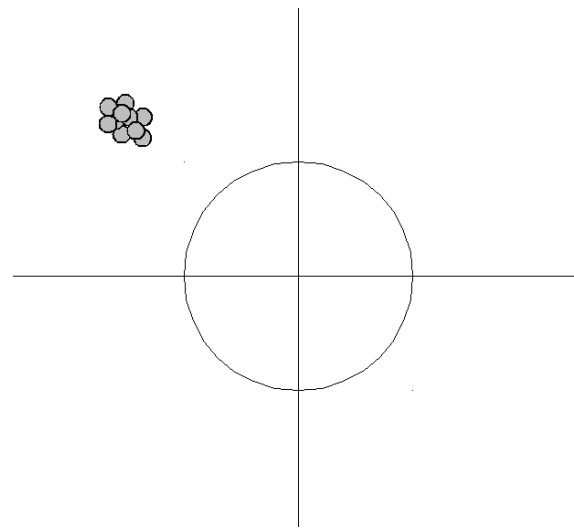
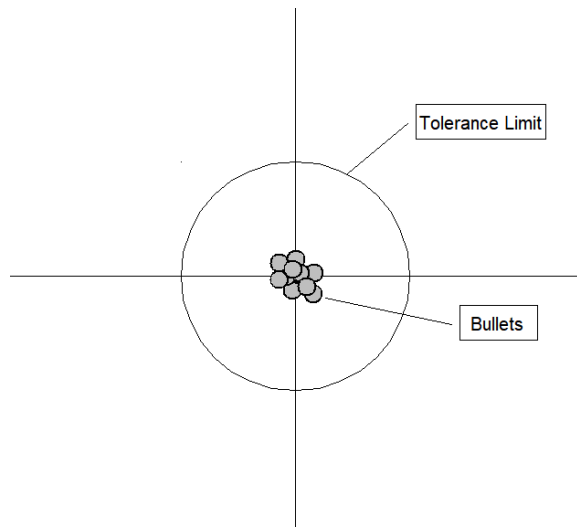
$$\text{bias} = \mathbb{E}_{x \sim \mathcal{D}_{\text{test}}} \left[\left(\bar{f}_\theta(x) - f(x) \right)^2 \right].$$

- **variance** denotes the variability in between the individual training runs:

$$\text{variance} = \mathbb{E}_{x \sim \mathcal{D}_{\text{test}}} \left[\mathbb{E}_{\mathcal{D} \sim \{\mathcal{D}_{\text{train},\ell}\}_{\ell=1}^{\infty}} \left[\left(f_\theta^{(\mathcal{D})}(x) - \bar{f}_\theta(x) \right)^2 \right] \right].$$

⇒ Often a matter of model complexity.



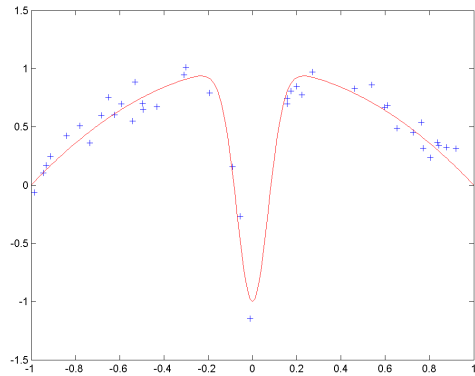


Bias-variance tradeoff (2)

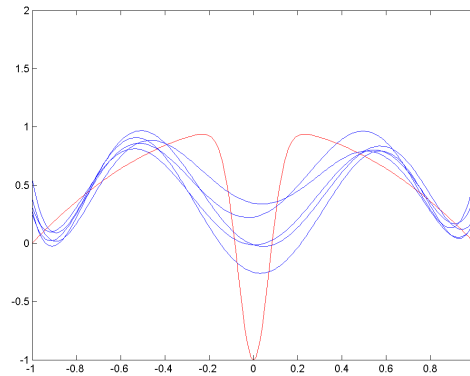
Example (see [Wikipedia](#)): Fitting a model of several radial basis functions to noisy training data:

$$f_{\theta}(x) = \sum_{k=1}^d \theta_k \exp \left(-\frac{1}{2} \frac{x - c_k}{\sigma_k^2} \right).$$

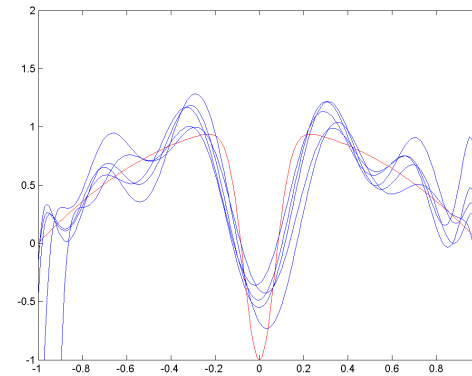
- For a wide spread (i.e., large σ_k), the bias is high: the RBFs cannot fully approximate the function (especially the central dip), but the variance between different trials is low.
- As spread decreases (image 3 and 4) the bias decreases: the blue curves more closely approximate the red...
- ... but the variance between trials ($\mathcal{D}_{\text{train},1}, \mathcal{D}_{\text{train},2}, \dots$) increases.



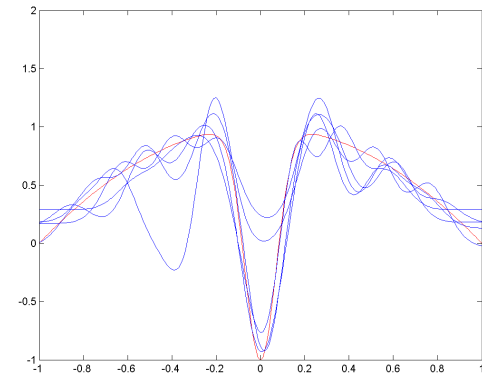
Function and training data $\mathcal{D}_{\text{train},1}$.



Wide spread RBFs.

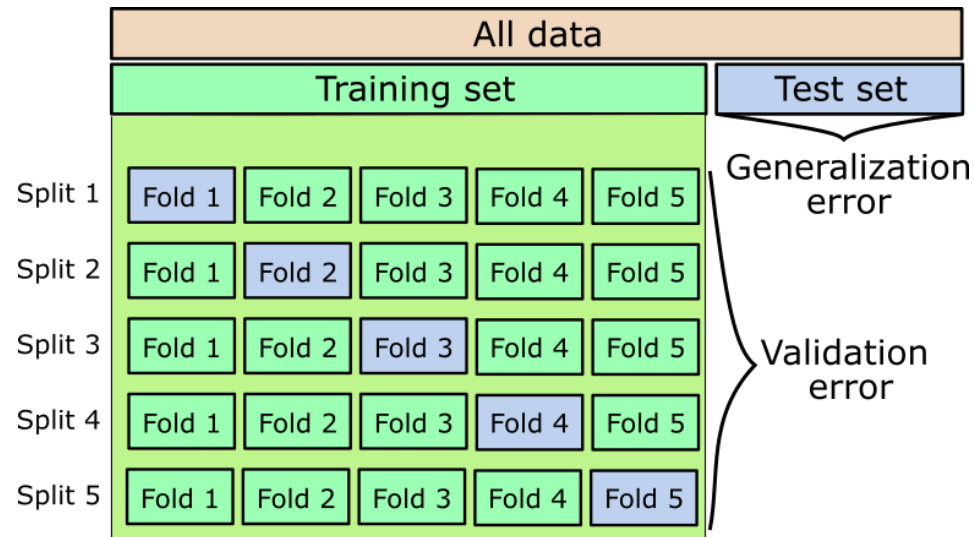


Medium spread RBFs.



Small spread RBFs.

Cross-validation

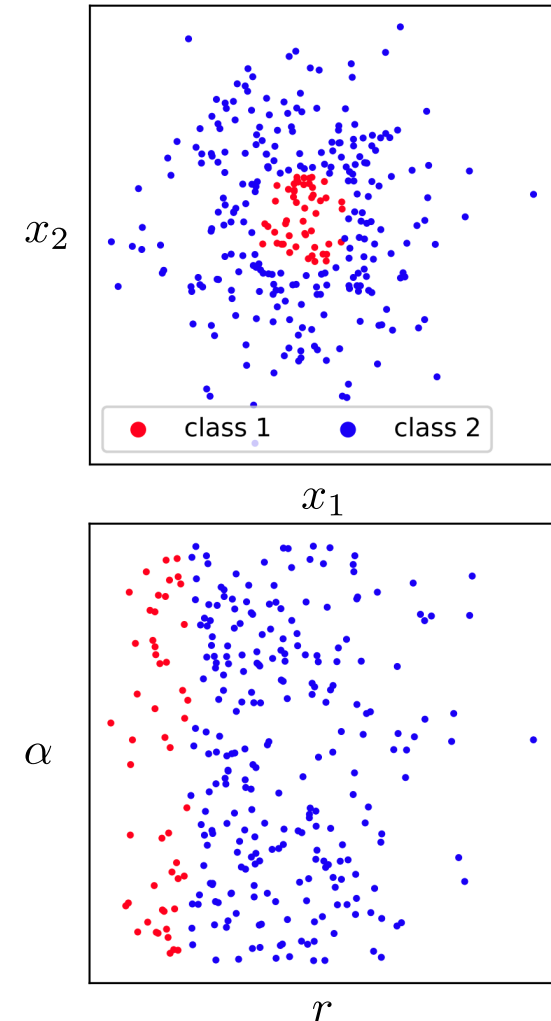


5-fold CV example ([Abdelwanis et al. 2026](#)).

- Training is repeated k times with k different splits of the training set.
- Each observation serves as unseen instance (blue boxes) at least once.
- The validation error is an indicator for tuning hyperparameters.
- Example of a k -fold Cross-validation (CV).

Means to improve a supervised learning model

- Collecting **more data**, i.e., increasing N .
- **Reducing noise** in the data.
- **Improving the distribution** within the dataset, i.e., ensuring that the data set is representative of the problem domain.
- **Choosing a more appropriate model**. A general rule is to select the model according to the amount of data one has, not according to the expected complexity of the function to approximate.
- **Optimizing hyperparameters** of the model.
- **Ensemble learning**: Averaging over several different models.
- **Including knowledge**, e.g., in the form of tailored features (feature engineering) or informed loss functions.
 - This is known as **inductive bias** in the ML literature.



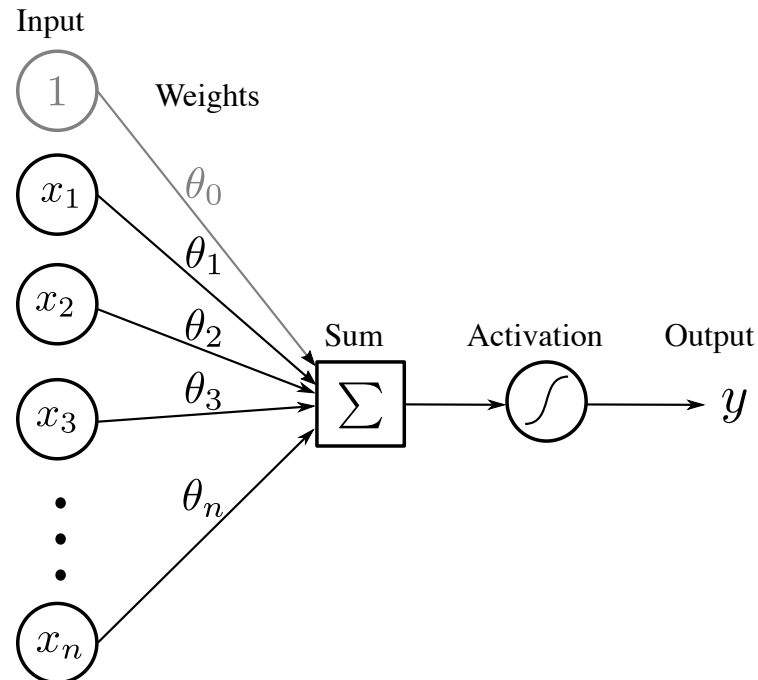
Euclidean vs. polar coordinates/features in binary classification (Abdelwanis et al. 2026).

Deep neural networks

Artificial neural networks

Artificial neural networks (ANNs) are nonlinear function approximators $\hat{y} = f_{\theta}(x)$ that

- are end-to-end differentiable.
- are stacks of minimal units, the **artificial neurons**.



An artificial neuron (Abdelwanis et al. 2026).

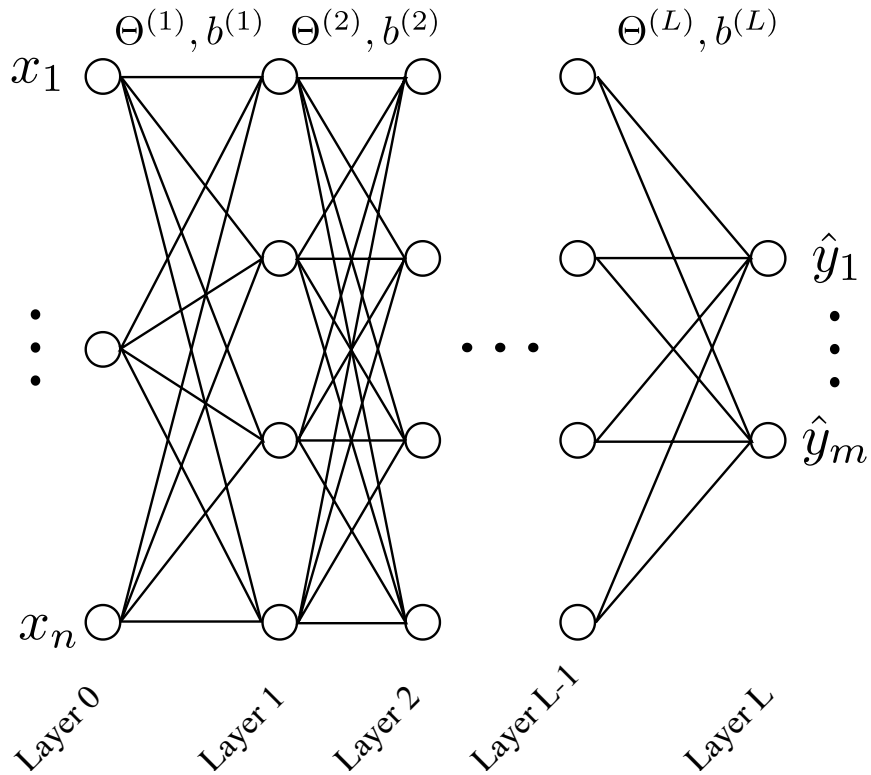
- An ANN consists of **nodes** or **neurons** in one or more layers.
- Each node transforms the weighted sum of all previous nodes (plus a potential **bias term**) through an **activation function** σ :

$$\sigma \left(\theta_0 + \sum_{k=1}^n \theta_k x_k \right).$$

- The weighted connections are called **edges**, which represent the ANN's parameters.

Multi-layer perceptron

Standard model of supervised learning: **multi-layer perceptron** or **feed-forward ANN**.



MLP architecture (Abdelwanis et al. 2026).

- Only forward-flowing edges.
- The depth L and width $H^{(\ell)}$ are hyperparameters.
- With $\sigma^{(\ell)}$ and $z^{(\ell)}$ denoting the activation function and **activation** of layer ℓ respectively, we get for the output in the ℓ^{th} layer.

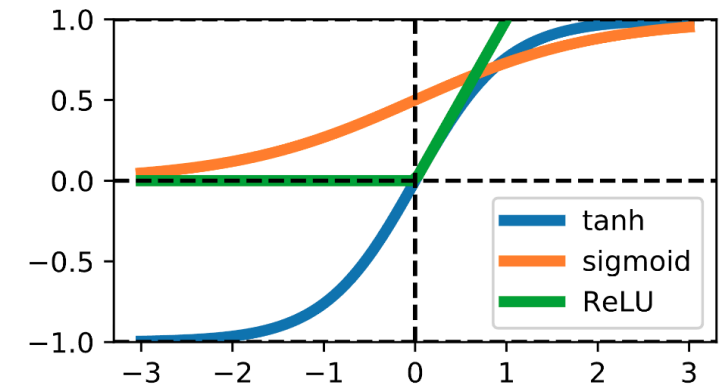
$$x^{(\ell)} = \sigma^{(\ell)} \left(\underbrace{\Theta^{(\ell)} x^{(\ell-1)} + b^{(\ell)}}_{z^{(\ell)}} \right),$$

with input $x^{(0)} = x$ and output $x^{(L)} = y$:

- Training:
 - Summarize the full set of parameters (i.e., weight matrices $\Theta^{(\ell)} \in \mathbb{R}^{H^{(\ell)} \times H^{(\ell-1)}}$ and biases $b^{(\ell)}$) under θ .
 - Iteratively update the weights using gradient information.

Activation functions

- The source of nonlinearity in neural networks.*
- Common choices for $\sigma(z)$ are
 - $\sigma(z) = \tanh(z)$,
 - Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$,
 - Rectified linear unit (ReLU): $\sigma(z) = \max(0, z)$,
- The activation of the output layer, $\sigma^{(L)}(z)$, is task-dependent. For instance,
 - regression: $y = \sigma^{(L)}(z^{(L)}) = z^{(L)}$, i.e., $\sigma^{(L)} = \text{Id}$ is the identity mapping.
 - binary classification: sigmoid (i.e., probability), followed by a rounding step to either 0 or 1.
 - multi-class classification: $y_i = \frac{\exp(z^{(L)}_i)}{\sum_j \exp(z^{(L)}_j)}$ (softmax).



Exemplary activation functions (Abdelwanis et al. 2026).

* Without nonlinear activation functions, every ANN collapses to a single matrix-vector multiplication $y = \hat{\Theta}x$:

$$x^{(\ell+2)} = \Theta^{(\ell+2)} x^{(\ell+1)} = \Theta^{(\ell+2)} (\Theta^{(\ell+1)} x^{(\ell)}) = (\Theta^{(\ell+2)} \Theta^{(\ell+1)}) x^{(\ell)} = \hat{\Theta} x^{(\ell)}.$$

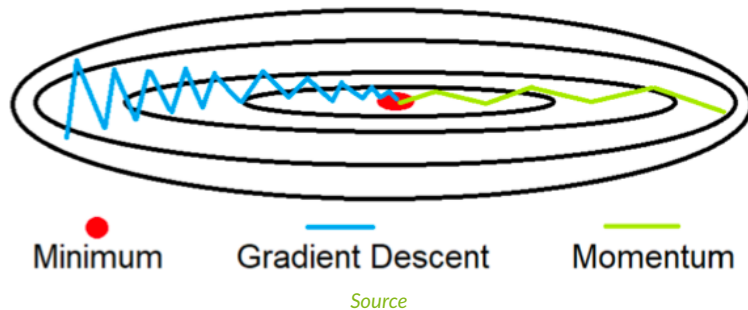
For non-zero biases, we obtain an *affine* transformation instead.

Training (1)

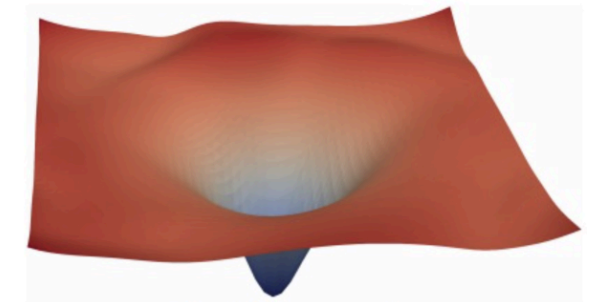
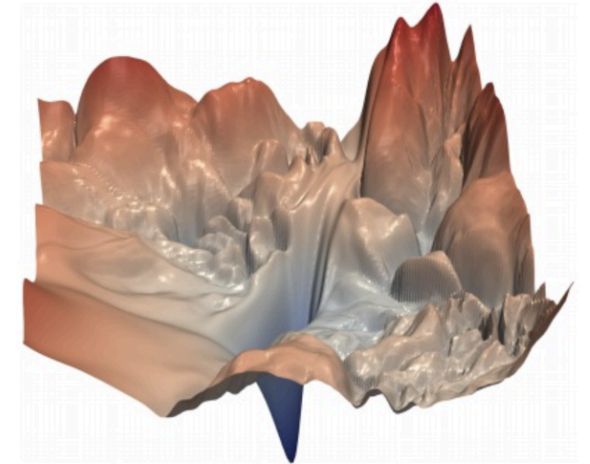
- Training is performed in an iterative manner:

$$\theta \leftarrow \theta + \eta \delta \theta.$$

- $\eta \in \mathbb{R}_{>0}$ is the **step size** or **learning rate**.
- $\delta \theta \in \mathbb{R}^d$ is the **update direction**, usually a gradient-based **descent direction**.
- Numerous variants for $\delta \theta$. The strongest contain additional **momentum** terms such as the *Adam* algorithm (Kingma and Ba 2014). We remember the previous update $\delta \theta^-$ and thereby flatten out zig-zag behavior.



- First, we need to define a **loss function** that we wish to minimize.
 - Regression: (root) mean square error, mean absolute error.
 - Classification: cross entropy.
 - Additional terms, e.g., regularization, physics information, ...
- Iterations over the dataset $\mathcal{D}_{\text{train}}$ are called **epochs**.



The loss landscape of deep neural networks (with and without skip-connection) (Li et al. 2018).

Training (2)

- The descent direction is computed by taking the derivative of the loss function w.r.t. the weight vector. In terms of the MSE:

$$\begin{aligned}\nabla L(\theta) &= \nabla \left(\frac{1}{N} \sum_{k=1}^N \|f_{\theta}(x_k) - y_k\|_2^2 \right) \\ &= \frac{1}{N} \sum_{k=1}^N \nabla \|f_{\theta}(x_k) - y_k\|_2^2 && \text{(Linearity of the sum)} \\ &= \frac{1}{N} \sum_{k=1}^N (2\|f_{\theta}(x_k) - y_k\|_2 \nabla f_{\theta}(x_k)) && \text{(Chain rule of differentiation)}\end{aligned}$$

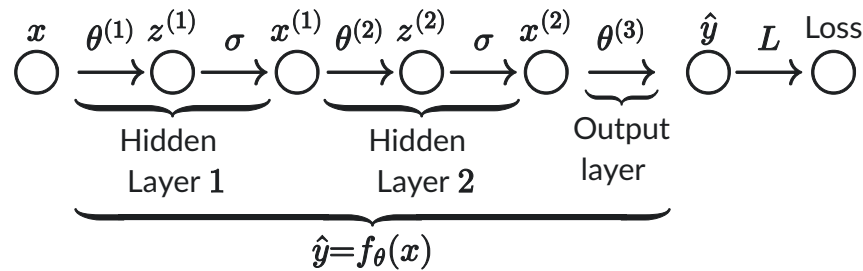
- This means that we need to *propagate* the error through our model f_{θ} .
- Since a neural network is a chain of neurons, we need to apply the **chain rule** of differentiation over the layers.
- As a consequence the loss is **backpropagated** through the network to determine the individual descent directions: $\frac{\partial L}{\partial \theta_i}$.

This is called the **backpropagation** algorithm. We require one forward pass and one backward pass for every data tuple (x_k, y_k) . Taking the average gives us the average steepest-descent improvement over the dataset $\mathcal{D}_{\text{train}}$ for the current θ .

Backpropagation example

Let's consider a very simple network with $x, y \in \mathbb{R}$ and two hidden layers with a single neuron each.

Symbolic



In mathematical terms

$$\begin{aligned}\hat{y} &= \theta^{(3)}(x^{(2)}) = \theta^{(3)}(\sigma(z^{(2)})) = \theta^{(3)}(\sigma(\theta^{(2)}(x^{(1)}))) \\ &= \theta^{(3)}(\sigma(\theta^{(2)}(\theta(z^{(1)})))) = \theta^{(3)}(\underbrace{\sigma(\theta^{(2)}(\theta^{(1)}x))}_{\text{Hidden Layer 2}})\end{aligned}$$

Let's assume a single sample (x, y) and the MSE loss: $L(\theta) = (\hat{y} - y)^2$. Using the chain rule, we get:

1. Gradient w.r.t. θ_3 :

$$\frac{\partial L}{\partial \theta_3} = \underbrace{\frac{\partial L}{\partial \hat{y}}}_{=2(\hat{y}-y)} \frac{\partial \hat{y}}{\partial \theta_3}.$$

2. Gradient w.r.t. θ_2 :

$$\frac{\partial L}{\partial \theta_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial x^{(2)}} \underbrace{\frac{\partial x^{(2)}}{\partial z^{(2)}}}_{=\sigma'} \frac{\partial z^{(2)}}{\partial \theta_2}.$$

3. Gradient w.r.t. θ_1 :

$$\frac{\partial L}{\partial \theta_3} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial x^{(2)}} \underbrace{\frac{\partial x^{(2)}}{\partial z^{(2)}}}_{=\sigma'} \frac{\partial z^{(2)}}{\partial x^{(1)}} \underbrace{\frac{\partial x^{(1)}}{\partial z^{(1)}}}_{=\sigma'} \frac{\partial z^{(1)}}{\partial \theta_1}.$$

$\Rightarrow$ we propagate the loss **back** through the network, reusing most of the previous calculations!

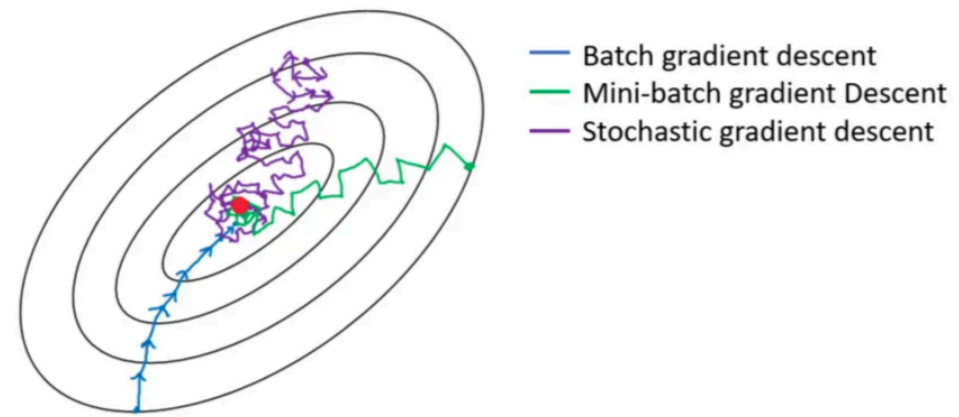
Stochastic gradient descent and batch learning

- The cost for a single gradient step scales with the dataset size N , which can be **very large**.
- For each sample, we need to perform one forward and one backward pass.
- **Stochastic gradient descent (SGD)**: massive speedup by using a single sample per step instead of the entire dataset,

$$\delta\theta = \nabla L(\theta; x_k), \quad k \sim U(\{1, \dots, N\}).$$

- **Mini-batch gradient descent** with $s \in \{1, \dots, N\}$ samples per step: compromise ground between efficiency and noisiness,

$$\delta\theta = \frac{1}{s} \sum_{k \in \mathcal{J}} \nabla L(\theta; x_k), \quad \text{where } \mathcal{J} \text{ is an } s\text{-dimensional, randomly drawn subset of } \{1, \dots, N\}.$$



Gradient descent vs. SGD vs. Mini-batch gradient descent [[Source](#)].

Regularization

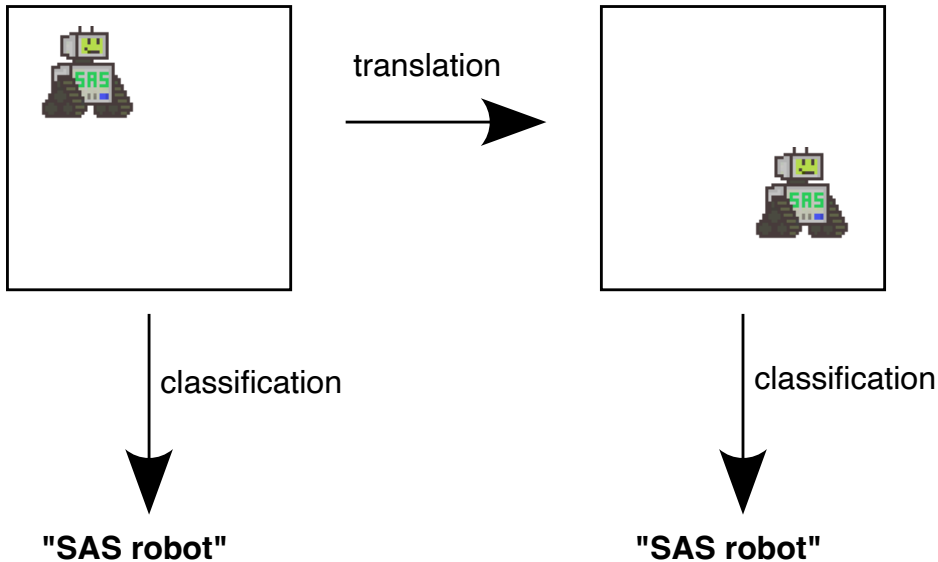
In order to mitigate overfitting (i.e., good performance on $\mathcal{D}_{\text{train}}$, poor generalization), neural networks can be regularized by

- **Weight decay:** adding an ℓ_2 penalty term to the weights: $L(\theta) + \lambda \|\theta\|_2^2$
- **Layer normalization** during training: all layers' activations are normalized separately by standard scaling,
- **Dropout:** randomly disable nodes' contribution.
 - This helps especially in deep networks,
 - and effectively builds an ensemble of ANNs with shared edges.

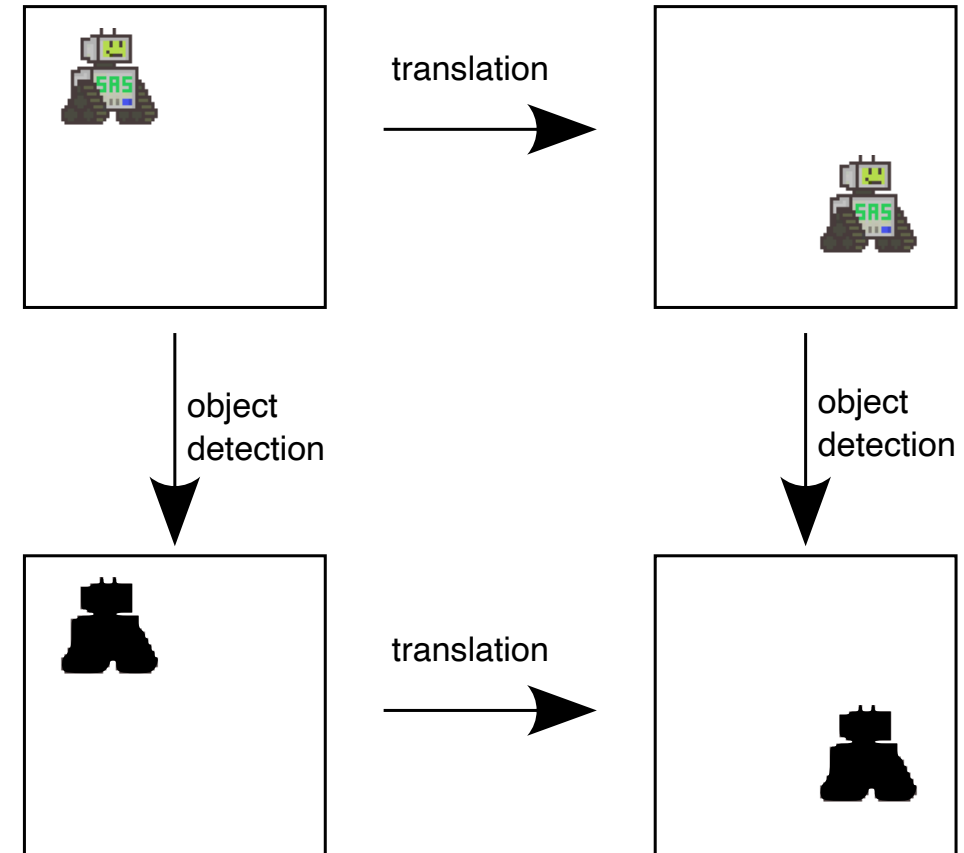
Convolutional neural networks

In machine learning (as well as in nature) **many tasks are independent of absolute position.**

A function is **invariant** under some transformation if its output y remains unchanged under transformations of the input x .



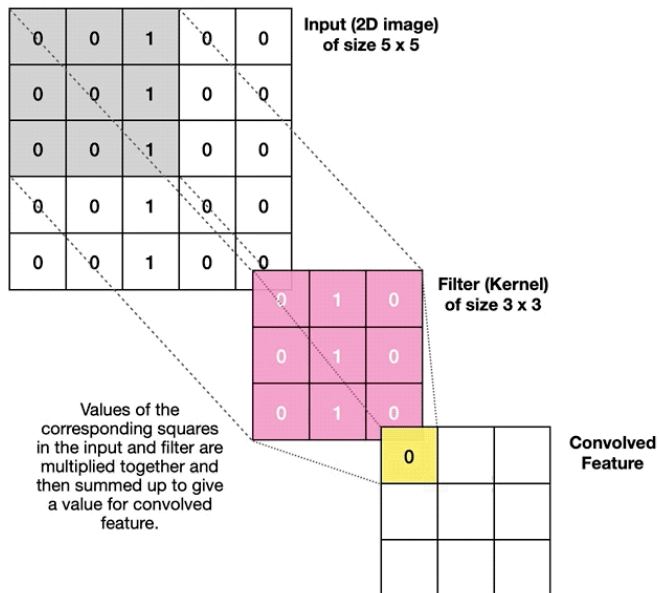
A function is **equivariant** under some transformation if its output y is transformed "in the same way" as the input x .



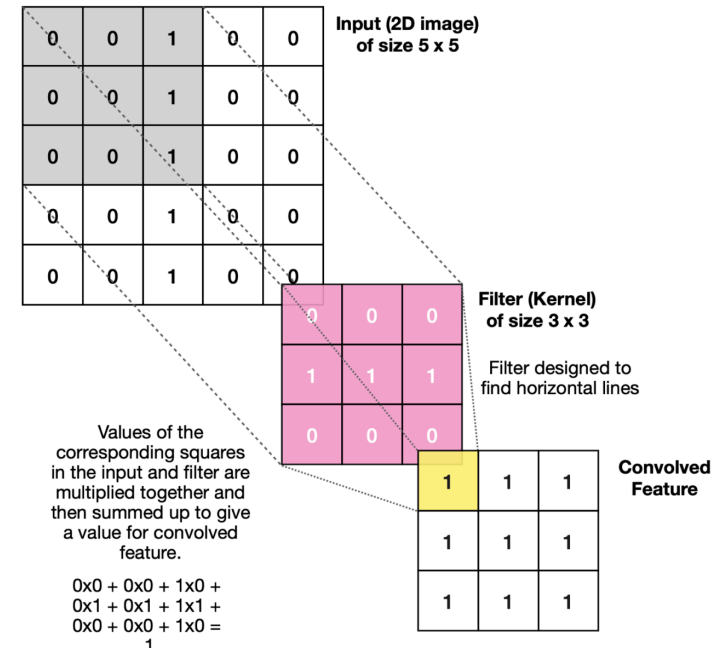
💡 These ideas can be formalized and extended under the framework of *group theory*, see also *geometric deep learning* (Bronstein et al. 2021).

Weight sharing in CNNs

- What does this mean for learning?
 - ⇒ We have to learn the same thing in different locations!
 - ⇒ An “edge” remains an “edge”, no matter where we are.
- Instead of training a fully connected layer, we train the weights of a **kernel** that moves over the input
- Same weights in every location ⇒ weight sharing!



- Architecture is closely related to fully connected NNs:
 - Inner product between kernel weights θ and input $x \Rightarrow$ activation z .
 - Multiple kernels $\Rightarrow$ multiple **channels**.
 - Then comes a nonlinear activation function $\sigma(z)$.
 - Further followed by additional steps (see next slide).

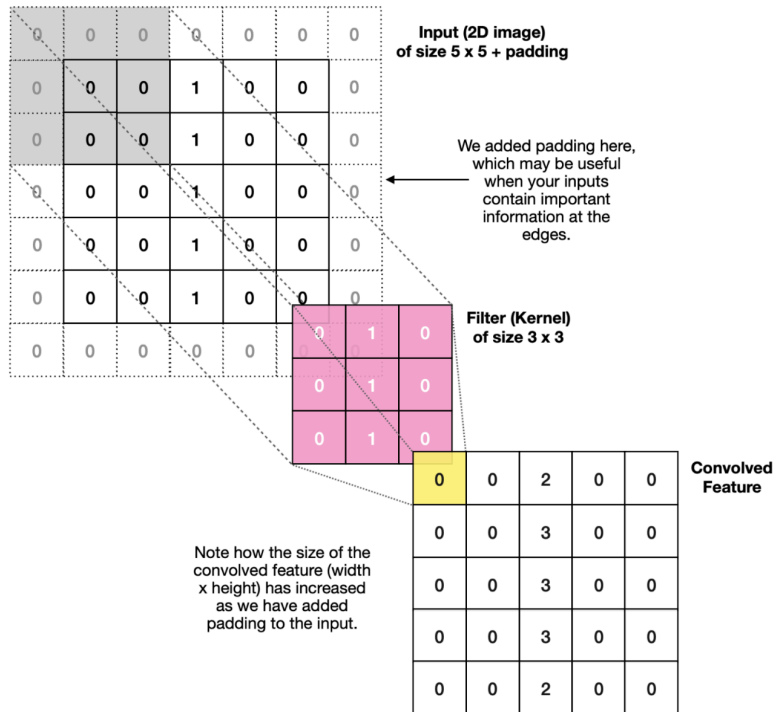


A kernel sweeping over an input [Source]

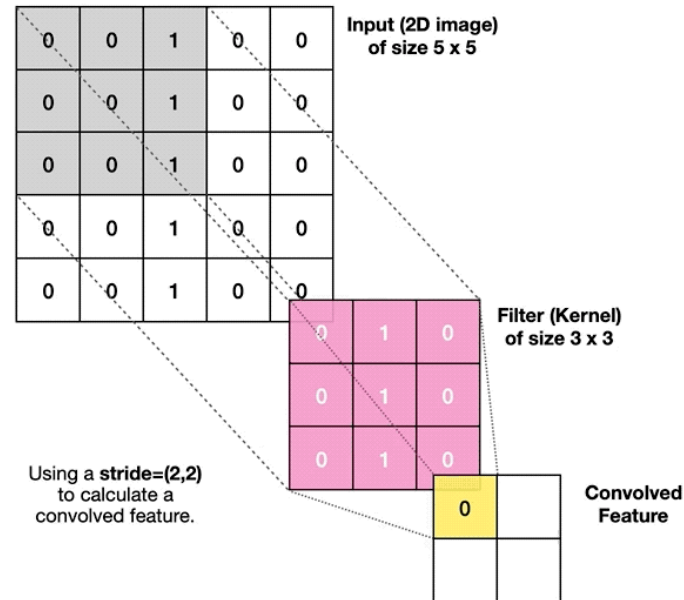
Second kernel = second channel [Source]

Ingredients of a CNN

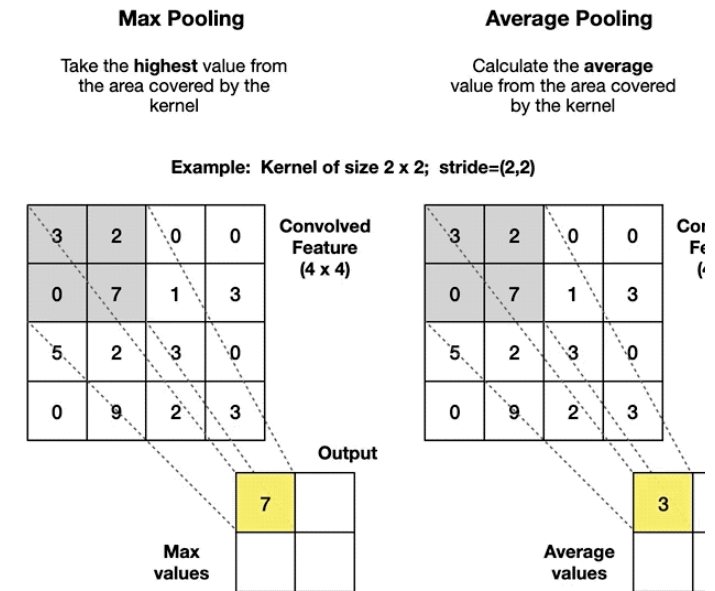
Further considerations (padding & stride) and additional steps (e.g., pooling)



Padding [Source]

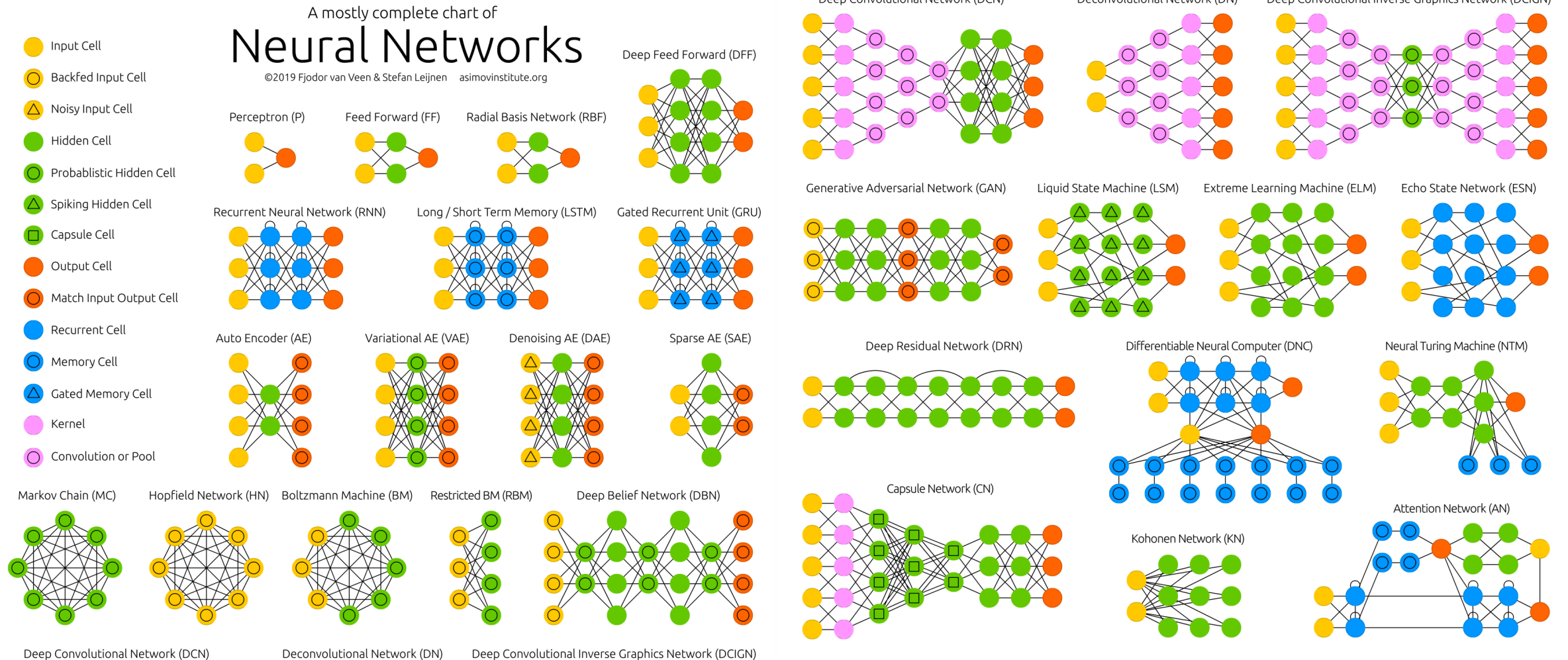


Stride [Source]



Pooling [Source]

Neural network architectures



The neural network zoo [Asimov Institute].

Linear regression as a special case

The linear model: single-layer NN with $\sigma = \text{Id}$

- Let's assume that we only have a single layer directly mapping inputs $x \in \mathbb{R}^n$ to outputs $y \in \mathbb{R}^m$:

$$\hat{y} = f_{\Theta}(x) = \Theta^{\top} x \quad \text{with} \quad \Theta \in \mathbb{R}^{m \times n}.$$

- Let's consider the usual MSE loss:

$$L(\Theta) = \frac{1}{N} \sum_{k=1}^N \|\Theta^{\top} x_k - y_k\|_2^2.$$

- We organize the data in a slightly different fashion, i.e.,

$$X = \begin{pmatrix} - & x_1^{\top} & - \\ & \vdots & \\ - & x_N^{\top} & - \end{pmatrix} \in \mathbb{R}^{N \times n}, \quad Y = \begin{pmatrix} - & y_1^{\top} & - \\ & \vdots & \\ - & y_N^{\top} & - \end{pmatrix} \in \mathbb{R}^{N \times m}.$$

- Then the model can predict all N samples at the same time by a matrix-matrix product: $\hat{Y} = X\Theta$.

- The associated loss function then simply is the **Frobenius norm** over the dataset $\mathcal{D}_{\text{train}} = (X, Y)$:

$$L(\Theta; \mathcal{D}_{\text{train}}) = \frac{1}{N} \|X\Theta - Y\|_F^2 = \frac{1}{N} (X\Theta - Y)^\top (X\Theta - Y).$$

Solving the regression problem

- What's the necessary condition for the minimizer of a function?
⇒ The gradient has to be zero!

$$\frac{\partial L(\Theta; \mathcal{D}_{\text{train}})}{\partial \Theta} = \frac{\partial}{\partial \Theta} \left[\frac{1}{N} (X\Theta - Y)^\top (X\Theta - Y) \right] = \frac{2}{N} X^\top (X\Theta - Y)$$

- Optimal solution:

$$\begin{aligned} \frac{2}{N} X^\top (X\Theta^* - Y) &\stackrel{!}{=} 0 \\ \Leftrightarrow X^\top X\Theta^* &= X^\top Y \\ \Leftrightarrow \Theta^* &= (X^\top X)^{-1} X^\top Y = X^\dagger Y, \end{aligned}$$

where $X^\dagger$ is the **pseudo-inverse** of X .

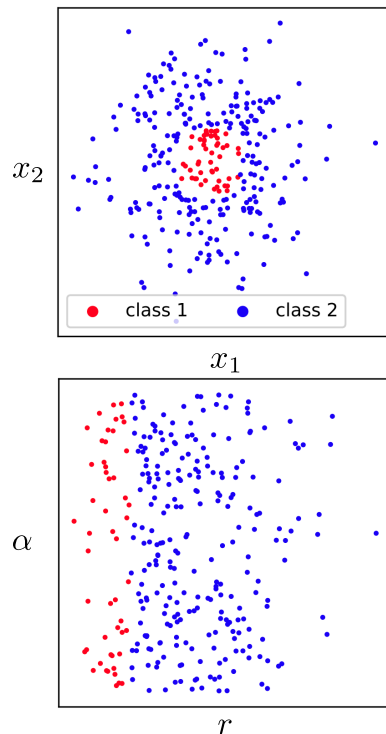
- Training is replaced by a simple one-step learning approach (using, e.g. `numpy.linalg.pinv`).
- Since the loss function L is convex, the solution Θ^* is not a local, but the **global optimum**.

Features as an additional pre-processing step

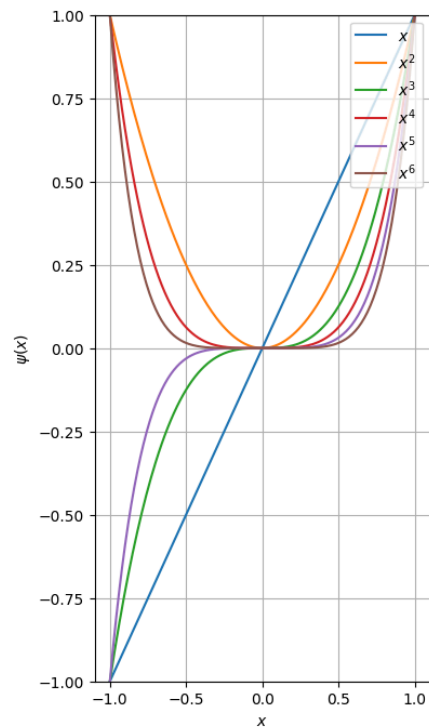
- If we introduce *feature functions* $\psi_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $\psi_i(x) = z_i$, then we can significantly improve the performance:

$$\psi(x) = [\psi_1(x), \dots, \psi_q(x)]^\top.$$

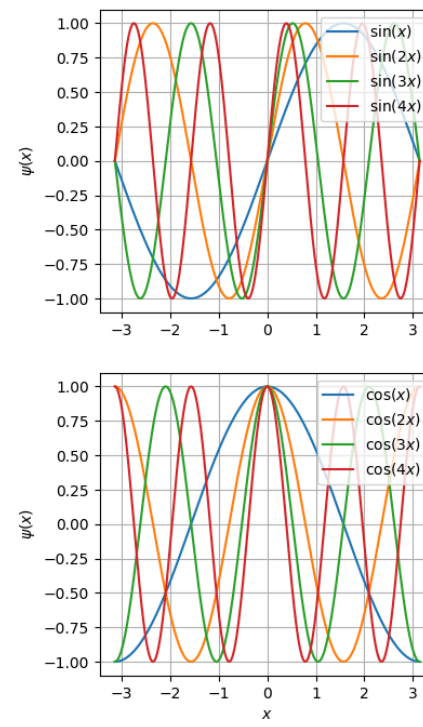
- Different ways of introducing features:



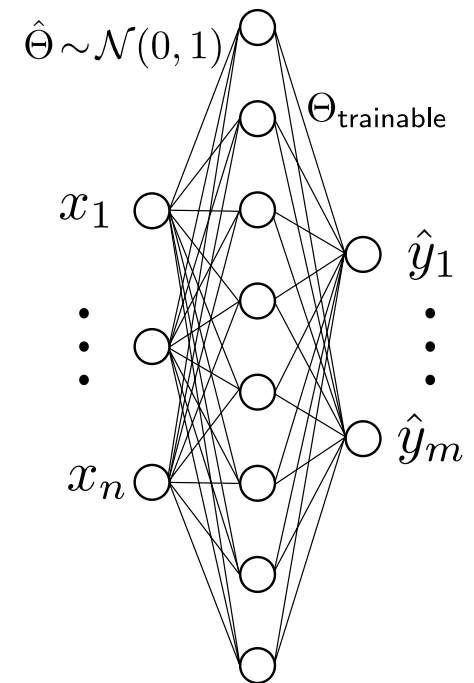
Tailored (Abdelwanis et al. 2026)



Manual: Polynomials



Manual: Fourier modes



Randomly: Extreme Learning Machine

💡 The layers 1 to $L - 1$ of a neural network can also be seen as a feature transform, **learned automatically** from data.

Summary / what you have learned

Summary / what you have learned

- There are **three main learning paradigms**:
 - **supervised learning** for function approximation of input-to-output mappings,
 - **unsupervised learning** for clustering, pattern detection or dimensionality reduction,
 - **reinforcement learning** for sequential decision making in dynamic environments.
- In supervised learning, we aim to approximate an unknown function f by a parametric function f_θ by adjusting θ such that a loss function L is minimized over a training dataset $\mathcal{D}_{\text{train}}$.
 - The central goal is **generalization** beyond the training data.
 - We have to handle the **bias-variance-tradeoff**, which is often a matter of model complexity.
 - There are many techniques to improve learning such as cross-validation, regularization, inductive biases, ...
- **Deep neural networks** come in a great range of varieties.
 - They are a large network of a single unit known as the perceptron.
 - The general structure consists of linear transformations, followed by nonlinear activation functions.
 - Training is realized via **backpropagation** and **gradient descent**.
 - Various versions for the descent direction, for instance using **momentum**.
 - Improved efficiency (at the cost of noise) by using **stochastic gradient descent** or **mini-batch gradient descent**.
- Linear regression is the special case of a single linear layer.
 - Training can be realized in closed form using the **pseudo-inverse**.

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